

ADVANTEST®

ADVANTEST CORPORATION

*ADVANsite Test System
Software Installation/
System Start User's Manual
VOL2 Tester Environment
Software Installation*

MANUAL NUMBER 8227729-21

Applicable Software

Solaris 8

Solaris 7

Solaris 2.5.1

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Preface

Purpose of This Manual

This manual explains the procedure from installation of the tester environment software to activation of the tester controller.

Related Manuals

ADVANsite Test System Software Installation/System Start User's Manual VOL1 Basic OS Installation(No. : 8227728)

H3-71171 UPS (Uninterruptible Power Supply) User's Manual(No. : 8292986)

T Series Maintenance Pack Rev 2.00 Release Note(No. : 8409633)

Command Reference Manual

1. Manual Overview and Cautions

This manual explains the procedure from installation of the tester environment software to activation of the tester controller.

For installing a basic OS software Solaris, follow [ADVANSite Test System Software Installation/System Start User's Manual VOL1 Basic OS Installation\(No. : 8227728\)](#).

When you are setting the tester environment, or only updating tester environment software for a system in which Solaris has already been installed, install the tester environment software using procedures described in this manual.

1. 1 Installation Type and Manual Configuration

There are two kinds of installation procedure: "quick installation" and "custom installation". Either procedure is selected to use in this case. Features of each procedure are shown as follows.

Quick installation

It is possible to select the multiple products stored in CD-ROM and install them in bulk. This way is very efficient in case of installing the multiple products to take time because input operations are concentrated right after installation script begin to start. Each kind of setting value used to be input by conventional way "asxinstall" are automatically set to be standard value.

➔ See [Table 1-2](#).

Custom installation

This way is used in case of installing by setting the value excepting the standard value that is automatically set in quick installation.

Specific procedure for operation of these installation are described in [2. "Quick Installation" on page 2-1](#) and [3. "Custom Installation" on page 3-1](#).

To use the tester, the following software should be installed.

Table 1-1 Software Required for the System

ASX/U type	Required Software
ASX/U-51 REV 5.05 and later	systest51su, testutl51su, diag****su, atw51su
Other ASX/U	systest**su, testutl**su, diag****su



Tip

*systest**su, testutl**su and diag****su vary according to the system model being used.*

In case of using quick installation, these products are selected and executed to install. If you use the custom installation method, install these software products individually.

Operation procedure on how to start the tester controller through how to set the each value are described in [4. "Installation of Data File for Tester Diagnosis" on page 4-1](#) and following chapters.

Table 1-2 Setting Items and Value Automatically by Simple Installation

Question items on conventional "asxinstall" (In case of simple installation way, there is no question on following item. Values are automatically set as shown in the right column.)	Setting value automatically by simple installation way.
Location of ASX system directory ADVANTEST softwares are installed under the root directory AT_ASPECIFIC. This script will create ADVANsite directories. The default directory is /export/home/atsys. Enter the directory name of AT_ASPECIFIC. * [/export/home/atsys] : ↵	Default value : /export/home/atsys
Selection of RELEASE, UPDATE, SUPPLEMENT (Only for testutl and diag) * Do you need RELEASE version [yes] ? ↵ * Do you need UPDATE version [yes] ? ↵ * Do you need SUPPLEMENT version [yes] ? ↵	Default value : yes

1. 2 Notes

Compatible environment

This manual covers Solaris 2.5.1, Solaris 7 and Solaris 8. Note however, that some products may not be compatible with Solaris 8, even if this manual is provided.

➔ For more information, refer to the appropriate "Update Notice" of your ASX/U.

Product name

"Product name" or "systest**su" is described in the input command to explain the installation in the next chapter and after. Substitute characters corresponding to a product to be installed for the "systest30su".

Installing systest**su

When you install systest**su, an I/F board should already be installed in the EWS (it is factory-installed).

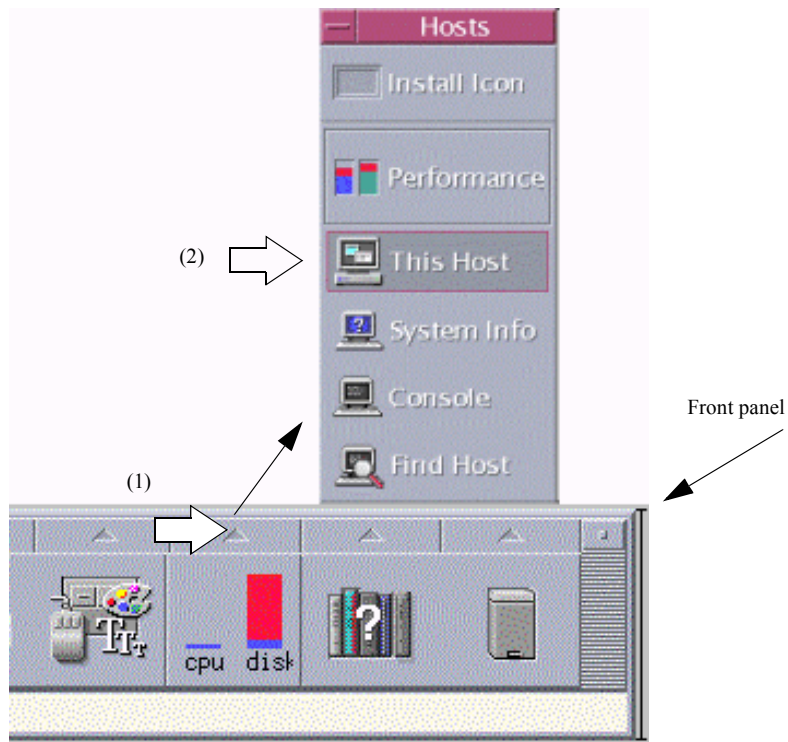
In addition, be sure to restart the system after "systest**su" has been installed.

CDE environment and terminal emulator startup

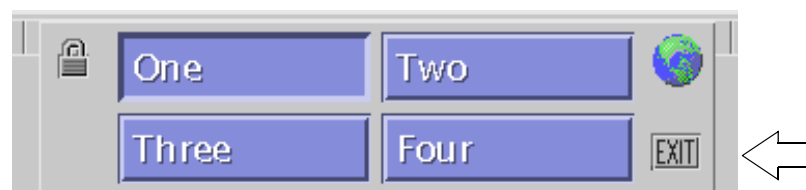
When a test system running on Solaris 7 and later is started, a login window (dtlogin) is displayed. Enter the user name and password to log in the system on Common Desktop Environment (CDE) as well as the conventional login.

To start the terminal emulator, click on (a) and (b) in [Figure 1-1](#).

Use this terminal emulator to enter a command in the command line as described in the following chapters.

Figure 1-1 CDE Front Panel (1/2)

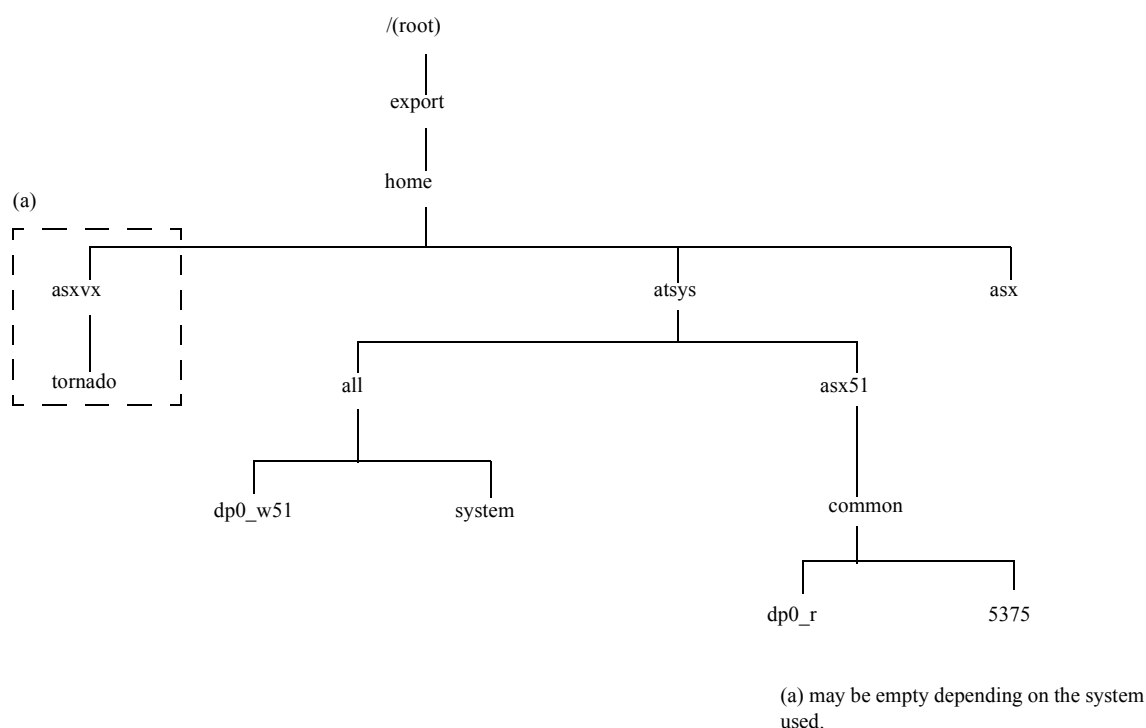
To log out the system, click EXIT of the front panel. A menu confirming the logout is displayed. To continue the logout, click OK.

Figure 1-2 CDE Front Panel (2/2)

1. 3 Directory Tree

The directory tree shown in [Figure 1-3](#) is built after ASX/U has been installed.

Figure 1-3 ASX/U Directory Tree (Example for a T5375 Test System)



Files used with ASX/U are saved in the directories below the directory `"/export/home"` as shown in [Figure 1-3](#).

`/export/home/asxvx/tornado`

Directory that includes the Tornado operating system used with the hybrid controller.

`/export/home/atsys`

Directories below "atsys" are as follows.

`all/dp0_w51`

Directory used to read or write dp0.

`all/system`

Directory used to store ASX/U application programs that are run on the environment processor.

`asx51/common/dp0_r`

Directory used to save files used with system utilities.

This directory is a read-only directory.

`asx51/common/5375`

Directory used to save diagnostic programs.

This directory is a read-only directory.

`/export/home/asx`

A user account with which the tester processor can be started.

2. Quick Installation

This chapter explains the quick installation.

! Important

Some products cannot be installed by this method. These products have the comment [Please install with "asxinstall"] in storing product display after the installation script starts. For the product which has this comment, install it according to [3. "Custom Installation" on page 3-1](#).

2. 1 Quick Installation

2. 1. 1 Install Script Starts from Local CD-ROM Drive

- 1) Log in with the "root" account.
- 2) Insert a CD-ROM containing programs into the drive.
- 3) Enter the following commands. Then the installation script is started.

```
# cd /cdrom/cdrom0␣  
# ./install.cmd␣
```

- 4) Go to [2. 1. 3 "Installation" on page 2-2](#).

2. 1. 2 Installation Script Starts from a Remote CD-ROM Drive

Setting up a remote system (a system connected the CD-ROM drive to)

➔ If the remote system is SunOS 4.x.x, refer to [Appendix 1. "Remote System Setup for SunOS4.x.x" on page A1-1](#).

- 1) Log in with the "root" account.
- 2) Insert a CD-ROM containing programs into the drive. The CD-ROM is mounted automatically.
- 3) Make sure that the CD-ROM has been mounted.

```
# ls /cdrom␣  
cdrom0  unnamed_cdrom
```

- 4) Set up to make the CD-ROM available to use in other systems.

```
# share -F nfs -o ro /cdrom/cdrom0
```

- 5) Start NFS mount demon. The following two commands does not have to be executed in a system set up as a desk server. Entering "ps -e" confirms whether the command was executed or not. If "nfsd" and "mountd" are displayed, the command does not have to be executed.

```
# /usr/lib/nfs/nfsd -a 8
# /usr/lib/nfs/mountd
```

Setting up a remote system is completed.

Setting up a local system (a system to be installed)

- 1) Log in with the "root" account.
- 2) Enter the following commands. Then the installation script is started.

```
# mkdir /cdrom
# mount remote node name: /cdrom/cdrom0 /cdrom
# cd /cdrom
# ./install.cmd
```

- 3) Go to [2. 1. 3 "Installation" on page 2-2](#).

2. 1. 3 Installation

- 1) After starting the installation script, product names included in a CD-ROM are displayed as shown below.

```
* The following product(s) are available:
  [1] ATW51SU                (Rev *.**)      ←This product name is displayed only
                                           for the appropriate systems.
  [2] DIAG****SU            (Rev *.**)
  [3] SYSTEST**SU           (Rev *.**)
  [4] TESTUTL**SU           (Rev *.**)

  [A] ALL install

  [Q] QUIT
```

 Tip

*The message [Please install with "asxinstall"] might be displayed beside [(Rev *.*)] display. The product which has this message cannot be installed by this quick installation script "install.cmd". When you want to install this product, install it according to the procedure described in [3. "Custom Installation" on page 3-1](#).*

- 2) Select a product number to be installed. Multiple numbers can be selected by using space between numbers. When "A" or "a" is selected, all products which can be installed are selected.

```
* Select product(s) you wish to process [n,n n,A,Q]: a↓
```

- 3) The system asks for the system name. Enter the system name.

 Tip

*This question is not asked if the system has already been set or product SYSTEM**SU is contained in the CD-ROM.*

```
* Select system name from following names.
  ASX/U-11E, ASX/U-12, ASX/U-30, ASX/U-50, ASX/U-51
  ASX/U-70, ASX/U-71, ASX/U-72, ASX/U-73, ASX/U-31
  ASX/U-80

* Enter the system name: ASX/U-**↓
```

- 4) The system asks for the series name. Enter the number corresponding the series name.

 Tip

*This question is not asked if the system has already been set or product DIAG****SU is contained in the CD-ROM.*

```
* Which series is this system?
  1. **** series
  2. **** series
  3. **** series

* Select number: *↓
```

- 5) Confirm that the products to be installed are correct. Entering y and the return key starts the installation.

 Tip

Confirmation lines of system name and series name are not displayed for the system which setting has already given to.

```
* Installation product(s):
  DIAG****SU
  SYSTEST**SU
  TESTUTL**SU
  *****

  ASX/U-** system
  T**** series
* Do you want to continue? [y,n]: y↵
```

- 6) The following message will be displayed to indicate that the installation is in progress.

```
////////// product install start //////////

Beginning installation of *****SU Rev *.* at **:**:*.
*****SU Rev *.* Installation completed successfully at **:**:*.

////////// product install end //////////
```

 Tip

- The following message might be displayed.
 - resetdev: *DEVICE is timeout Communications failure or loopback mode.*
 - resetdev: *TP or DEVICE is not ready.*

These messages are displayed if the TP or the NEW SDI unit are not powered on. However, installing is not influenced.
- If your test system uses Solaris 7, the following message may appear.
 - Installation of XXXXXX-xx failed. Return code 2.*
 - This message only appears when the patch has already been installed and has can be ignored.*

- 7) When the installation is completed, take out the media.

Execute the following commands.

Installing from a local CD-ROM	<pre># cd # eject cdrom</pre>	
Installing from a remote CD-ROM	When a system where a media is mounted is Solaris 2.x or Solaris 7 and later	When a system where a media is mounted is SunOS4.x.x
	- In a local system <pre># cd # umount /cdrom</pre> - In a system where a media is mounted (In a remote system) <pre># eject cdrom</pre>	- In a local system <pre># cd # umount /cdrom</pre> - In a system where a media is mounted (In a remote system) <pre># umount /cdrom/cdrom0 # eject cdrom</pre>

Take off the media.

8) Reboot the system.

Restart the system because some software packages require you to restart the system before they can be used.

Go to [4. "Installation of Data File for Tester Diagnosis" on page 4-1.](#)

3. Custom Installation

This chapter explains the custom installation.

3. 1 Installation (up to Starting the Installation Script)

3. 1. 1 Installing from a Local CD-ROM Drive

- 1) Log in with the "root" account.

```
hostname login: root.␣  
password:      ␣
```

- 2) Insert a CD-ROM containing programs into the drive.
- 3) Enter the following commands. Then the installation script is loaded to the disk.

```
# mkdir -p /export/home/atsupd.␣  
# cd /export/home/atsupd.␣  
# tar xf /cdrom/cdrom0/product-name.i.␣
```

- 4) Enter the following commands. Then the installation script is started.

```
# ./asxinstall product-name /cdrom/cdrom0.␣
```

- Refer to the description of each software product in [3. 2 "Installation \(Procedure after Starting the Installation Script\)" on page 3-3](#) for the concrete procedure after the installation script started.

3. 1. 2 Installing from a Remote CD-ROM Drive

Setting up a remote system (a system connected the CD-ROM drive to)

- If the remote system is SunOS 4.x.x, refer to [Appendix 1. "Remote System Setup for SunOS4.x.x" on page A1-1](#).

- 1) Log in with the "root" account.
- 2) Insert a CD-ROM containing programs into the drive. The CD-ROM is mounted automatically.

- 3) Make sure that the CD-ROM has been mounted.

```
# ls /cdrom␣  
cdrom0 unnamed_cdrom
```

- 4) Set up to make the CD-ROM available to use in other systems.

```
# share -F nfs -o ro /cdrom/cdrom0␣
```

- 5) Start NFS mount demon. The following two commands does not have to be executed in a system set up as a desk server.

Entering "ps -e" confirms whether the command was executed or not. If "nfsd" and "mountd" are displayed, the command does not have to be executed.

```
# /usr/lib/nfs/nfsd -a 8␣  
# /usr/lib/nfs/mountd␣
```

Setting up a remote system is completed.

Setting up a local system (a system to be installed)

- 1) Log in with the "root" account.
- 2) Enter the following commands. Then the installation script is loaded to the disk.

```
# mkdir /cdrom␣  
# mount remote node name: /cdrom/cdrom0 /cdrom␣  
# mkdir -p /export/home/atsetup␣  
# cd /export/home/atsetup␣  
# tar xf /cdrom/product-name.i␣
```

- 3) Enter the following commands. Then the installation script is started.

```
# ./asxinstall product-name /cdrom␣
```

- ➔ Refer to the description of each software product in [3. 2 "Installation \(Procedure after Starting the Installation Script\)" on page 3-3](#) for the concrete procedure after the installation script started.

3. 2 Installation (Procedure after Starting the Installation Script)

3. 2. 1 systest Installation

- 1) When starting the installation script "asxinstall" described in [3. 1 "Installation \(up to Starting the Installation Script\)" on page 3-1](#), the following message is displayed.

```
ADVANTEST Software Product Installation Script Rev x.xx
It is 18-JAN-1994 at 11:12:13.
```

- 2) If another user has logged in, the warning message is displayed and asks if you want to continue the installation.

If you continue, enter "yes" and the return key. To terminate installing, enter "no" and the return key, or just the return key.

```
The following processes are still active:
* Do you want to continue anyway [no] ? yes↵
```

- 3) The message asks you to create a backup copy of your system. Enter "yes" and the return key, or just the return key to continue installing. To terminate installing, enter "no" and the return key.

```
* Are you satisfied with the backup of your system disk [yes] ?↵
```

- 4) The message asks if you are ready for the installation. Enter "yes" and the return key to continue installing.

```
* Are you ready? yes↵
Beginning installation of SYSTEST**SU Rev x.xx at 11:12:13
Restoring product from tar file...
SYSTEST**SU Rev x.xx Installation is commencing.
```

The name and revision number of the program to be installed are displayed.

- 5) The message asks you to enter the ASX system directory at the prompt.

The default directory is set to /export/home/atsys. To use the default directory, press the return key.

3. 2 Installation (Procedure after Starting the Installation Script)

 Tip

This prompt message is not output if the system directory has already set.

```
ADVANSite softwares are installed under the root directory
AT_ASYSROOT. This script will create ADVANSite directories.
The default directory is /home/atsys.
Enter the directory name of AT_ASPECIFIC
* [/home/atsys ] :␣
```

- 6) Verify that the location is correct. Enter "yes" and the return key or just the return key, and the directory shown in Step 5) is created. Then control passes to the next step.

 Tip

This prompt message is not output if the system directory has already set.

```
Environment variable AT_ASPECIFIC is /home/atsys.
* Are you sure ? yes␣
```

- 7) The following message will be displayed to indicate that the installation is in progress.

```
Now copy the files to the target directories...
```

- 8) When the installation (file copy) is completed, the following message is displayed:

```
SYSTEST**SU Rev x.xx Installation completed successfully at 11:13:14.
```

 Tip

- *The following message might be displayed.*
 - resetdev: *DEVICE is timeout Communications failure or loopback mode.*
 - resetdev: *TP or DEVICE is not ready.*

These messages are displayed if the TP or the NEW SDI unit are not powered on. However, installing is not influenced.
- *If your test system uses Solaris 7, the following message may appear.*

Installation of XXXXXX-xx failed. Return code 2.

3. 2 Installation (Procedure after Starting the Installation Script)

This message only appears when the patch has already been installed and has can be ignored.

Execute the installation of testutl**su, successively.

3. 2. 2 testutl Installation

- 1) When starting the installation script "asxinstall" described in [3. 1 "Installation \(up to Starting the Installation Script\)" on page 3-1](#), the following message is displayed.

```
ADVANTEST Software Product Installation Script Rev x.xx  
It is 18-JAN-1994 at 11:12:13.
```

- 2) If another user has logged in, the warning message is displayed and asks if you want to continue the installation.

If you continue, enter "yes" and the return key. To terminate installing, enter "no" and the return key, or just the return key.

```
The following processes are still active:  
* Do you want to continue anyway [no] ? yes↵
```

- 3) The message asks you to create a backup copy of your system. Enter "yes" and the return key, or just the return key to continue installing. To terminate installing, enter "no" and the return key.

```
* Are you satisfied with the backup of your system disk [yes]?↵
```

- 4) The message asks if you are ready for the installation. Enter "yes" and the return key to continue installing.

```
* Are you ready? yes↵  
Beginning installation of TESTUTL**SU Rev x.xx at 11:12:13  
Restoring product from tar file...  
TESTUTL**SU Rev x.xx Installation is commencing.
```

The name and revision number of the program to be installed are displayed.

- 5) The message asks you to enter the ASX system directory at the prompt.

The default directory is set to /export/home/atsys. To use the default directory, press the return key.

3. 2 Installation (Procedure after Starting the Installation Script)

 Tip

This prompt message is not output if the system directory has already set.

```
ADVANSite softwares are installed under the root directory
AT_ASYSROOT. This script will create ADVANSite directories.
The default directory is /home/atsys.
Enter the directory name of AT_ASPECIFIC
* [/home/atsys ] :␣
```

- 6) Verify that the location is correct. Enter "yes" and the return key or just the return key, and the directory shown in Step 5) is created. Then control passes to the next step.

 Tip

This prompt message is not output if the system directory has already set.

```
Environment variable AT_ASPECIFIC is /home/atsys.
* Are you sure ? yes␣
```

7) Selection of RELEASE, UPDATE, SUPPLEMENT.

```
* Do you need RELEASE version [yes] ? ␣
* Do you need UPDATE version [yes] ? ␣
* Do you need SUPPLEMENT version [yes] ? ␣
```

In the above setting, selected versions are all installed.

Since the selected versions are installed in order of "RELEASE", "UPDATE", and "SUPPLEMENT", the latest version can be easily installed by simply entering "yes" and the return key or just the return key for each request.

The differences of these versions are as follows.

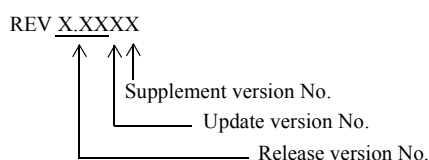
- Release version
Contains basic system functions and is the base of the other versions.
- Update version
Supports the release version and contains improved faults and additional functions.
- Supplement version

3. 2 Installation (Procedure after Starting the Installation Script)

Supports the release and update versions, and is issued when faults must be urgently improved or when functions must be urgently added.

When the latest release version is issued, the update and supplement versions are also issued at the same time. The update and supplement versions does not have to be installed at that time since the latest release version contains the former update and supplement versions. However, these three versions should be all installed because installing all three versions prevents to forget to install the latest update and supplement versions supporting the former release version when they are issued.

The revision number consists of each version as follows.



8) The following message will be displayed to indicate that the installation is in progress.

```
Restoring product from tar file...
Restoring product from tar file...
Restoring product from tar file...
    Now copy the files to the target directories...
```

The version selected with "yes" in 7) is displayed.

```
ADVANTEST Software Product Installation Script Rev x.xx
It is 18-JAN-1994 at 11:12:13.
```

Execute the installation of diag***su, successively.

3. 2. 3 diag Installation

1) When starting the installation script "asxinstall" described in [3. 1 "Installation \(up to Starting the Installation Script\)" on page 3-1](#), the following message is displayed.

ADVANTEST Software Product Installation Script Rev x.xx
It is 18-JAN-1994 at 11:12:13.

- 2) If another user has logged in, the warning message is displayed and asks if you want to continue the installation.

3. 2 Installation (Procedure after Starting the Installation Script)

If you continue, enter "yes" and the return key. To terminate installing, enter "no" and the return key, or just the return key.

```
The following processes are still active:
* Do you want to continue anyway [no] ? yes␣
```

- 3) The message asks you to create a backup copy of your system. Enter "yes" and the return key, or just the return key to continue installing. To terminate installing, enter "no" and the return key.

```
* Are you satisfied with the backup of your system disk [yes] ?␣
```

- 4) The message asks if you are ready for the installation. Enter "yes" and the return key to continue installing.

```
* Are you ready? yes␣
Beginning installation of DIAG****SU Rev x.xx at 11:12:13
Restoring product from tar file...
DIAG****SU Rev x.xx Installation is commencing.
```

The name and revision number of the program to be installed are displayed.

- 5) The message asks you to enter the ASX system directory at the prompt.

The default directory is set to /export/home/atsys. To use the default directory, press the return key.



This prompt message is not output if the system directory has already set.

```
ADVANSite softwares are installed under the root directory
AT_ASYSROOT. This script will create ADVANSite directories.
The default directory is /home/atsys.
Enter the directory name of AT_ASPECIFIC
* [/home/atsys ] :␣
```

- 6) Verify that the location is correct. Enter "yes" and the return key or just the return key, and the directory shown in Step 5) is created. Then control passes to the next step.

3. 2 Installation (Procedure after Starting the Installation Script)

 Tip

This prompt message is not output if the system directory has already set.

```
Environment variable AT_ASPECIFIC is /home/atSYS.  
* Are you sure ? yes↵
```

7) Selection of RELEASE, UPDATE, SUPPLEMENT.

```
* Do you need RELEASE version [yes] ? ↵  
* Do you need UPDATE version [yes] ? ↵  
* Do you need SUPPLEMENT version [yes] ? ↵
```

In the above setting, selected versions are all installed.

Since the selected versions are installed in order of "RELEASE", "UPDATE", and "SUPPLEMENT", the latest version can be easily installed by simply entering "yes" and the return key or just the return key for each request.

The differences of these versions are as follows.

- Release version

Contains basic system functions and is the base of the other versions.

- Update version

Supports the release version and contains improved faults and additional functions.

- Supplement version

Supports the release and update versions, and is issued when faults must be urgently improved or when functions must be urgently added.

When the latest release version is issued, the update and supplement versions are also issued at the same time. The update and supplement versions does not have to be installed at that time since the latest release version contains the former update and supplement versions. However, these three versions should be all installed because installing all three versions prevents to forget to install the latest update and supplement versions supporting the former release version when they are issued.

3. 2 Installation (Procedure after Starting the Installation Script)

Restart the system because some software packages require you to restart the system before they can be used.

Go to [4. "Installation of Data File for Tester Diagnosis" on page 4-1.](#)

3. 2. 4 ATLworks Installation

- 1) When starting the installation script "asxinstall" described in [3. 1 "Installation \(up to Starting the Installation Script\)" on page 3-1](#), the following message is displayed.

```
ADVANTEST Software Product Installation Script Rev x.xx  
It is 18-JAN-1994 at 11:12:13.
```

- 2) If another user has logged in, the warning message is displayed and asks if you want to continue the installation.

If you continue, enter "yes" and the return key. To terminate installing, enter "no" and the return key, or just the return key.

```
The following processes are still active:  
* Do you want to continue anyway [no] ? yes↵
```

- 3) The message asks you to create a backup copy of your system. Enter "yes" and the return key, or just the return key to continue installing. To terminate installing, enter "no" and the return key.

```
* Are you satisfied with the backup of your system disk [yes] ?↵
```

- 4) The message asks if you are ready for the installation. Enter "yes" and the return key to continue installing.

```
* Are you ready? yes↵  
Beginning installation of ATW51SU Rev x.xx at 11:12:13  
Restoring product from tar file...  
ATW51SU Rev 1.00 Installation is commencing.
```

The name and revision number of the program to be installed are displayed.

- 5) The message asks you to enter the ASX system directory at the prompt.

The default directory is set to /export/home/atSYS. To use the default directory, press the return key.

3. 2 Installation (Procedure after Starting the Installation Script)

Tip

This prompt message is not output if the system directory has already set.

```
ADVANTEST software are installed under the root directory
AT_ASPECIFIC. This script will create ADVANsite directories.
The default directory is /export/home/atsys.

Enter the directory name of AT ASPECIFIC
* [/export/home/atsys.] :↵
```

- 6) Verify that the location is correct. Enter "yes" and the return key or just the return key, and the directory shown in Step 5) is created. Then control passes to the next step.

Tip

This prompt message is not output if the system directory has already set.

```
Environment variable AT_ASPECIFIC is /home/atsys.
* Are you sure ? yes.↵
```

- 7) The following message will be displayed to indicate that the installation is in progress.

```
Now copy the files to the target directories...
```

- 8) The following message is displayed after the installation.

```
ATW51SU Rev 1.00 Installation completed successfully at 11:13:14 .
asxuinstall script done at 11:22:11
```

- 9) When the installation is completed, take out the media.

 3. 2 Installation (Procedure after Starting the Installation Script)

Execute the following commands.

Installing from a local CD-ROM	<pre># cd # eject cdrom</pre>	
Installing from a remote CD-ROM	When a system where a media is mounted is Solaris 2.x or Solaris 7 and later	When a system where a media is mounted is SunOS4.x.x
	- In a local system <pre># cd # umount /cdrom</pre> - In a system where a media is mounted (In a remote system) <pre># eject cdrom</pre>	- In a local system <pre># cd # umount /cdrom</pre> - In a system where a media is mounted (In a remote system) <pre># umount /cdrom/cdrom0 # eject cdrom</pre>

Take off the media.

10) Reboot the system.

Restart the system because some software packages require you to restart the system before they can be used.

Go to [4. "Installation of Data File for Tester Diagnosis" on page 4-1.](#)

4. Installation of Data File for Tester Diagnosis

To execute the test system diagnostic program, the system configuration file containing each boards revision information for the tester unit is necessary. When full initialization is made to a disk, the system configuration file should be installed. Install the system configuration program according to the procedure described in "Test System Tester Section Maintenance Manual" of your test system.

5. *System Start and Stop*

This chapter explains how to start and stop the environment processor and the tester processor (and test system).

5. 1 **System Start and Stop Overview**

The ADVANsite test system consists of the environment processor and the tester processor. In the current ADVANsite test system, it is possible to start/stop the environment processor and the tester processor (and tester system), separately.

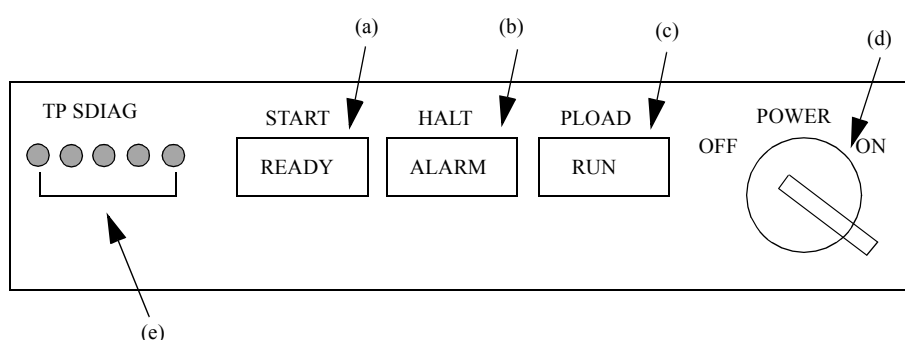
- ➔ For information on how to start and stop testers compatible with MOST and ASX/U (in other words, testers which use a single tester processor with an attached EWS), refer to [Appendix 6. "Testers Start and Stop for Both MOST and ASX/U" on page A6-1.](#)

5. 2 ASX/U Test System Start and Stop

5. 2. 1 Environment Processor Start

- 1) Push START switch ((a) in [Figure 5-1](#)) of AC control panel. READY lamp is on.

Figure 5-1 ASX/U Test System AC Control Panel



- 2) The environment processor is powered on by pushing START switch.

Tip

Depending on the tester processor used, the following message may be displayed on the console screen immediately after the environment processor starts up.

"WARNING: TP*: test processor not ready."

(4A, 5, or 6 is displayed for *, depending on the tester processor type.)

It is because the power of the tester processor is off. It has no effect on operating the environment processor.

- 3) Many kinds of starting information are scrolled on the screen and finally the scrolled following display appears, and finally the following message is displayed.

```
hostname login:
```

If you use Solaris 7 and later, the login screen through GUI is then displayed.

- 4) The system start is then completed and the system enters the status where log in is enabled.

! Important

The following accounts are registered in ADVANsite at shipping.

Account	Password
root	None
asx	asx-asx

Set the password or change the password for security.

5. 2. 2 Tester Processor Start

Turn on the POWER switch ((d) of [Figure 5-1](#)) of AC control panel. ALARM lamp goes off and RUN lamp is on.

Check that LED displays of TP SDIAG ((e) of [Figure 5-1](#)) goes off at this time.

Test systems that use tester processor TP6 and later do not have LED for TP SDIAG.

Instead, the message "target ready" is displayed on the console screen of the environment processor.

5. 2. 3 Environment Processor Stop

- 1) Log in to root.
- 2) Enter the shutdown command.

```
# /usr/sbin/shutdown -y -i0 -g0
```

- 3) The ok prompt is shown on the screen and then the environment processor stops.

```
The system is down
syncing file systems... down
Program terminated
Type help for more information
ok
```

However, to power off the environment processor, shut down the main breaker of the test system or push EMO switch on the left front side of the system after the environment processor is stopped.

5. 2. 4 Tester Processor Stop

- 1) Push HALT switch ((b) in [Figure 5-1](#)) of AC control panel. ALARM lamp is on and RUN lamp goes off.

- 2) Turn off the POWER switch ((d) of [Figure 5-1](#)).

5. 2. 5 Emergency Stop

Push EMO switch of the left front side of the system.

! Important

When the above emergency stop is performed, the file system may be damaged. When the system is reactivated, the command fsck (file system check) recovers the damaged part automatically as much as possible. However, if the file system has been badly damaged, the command fsck might ask the operator who performs installation for details of the processing.

➔ For details, refer to the explanation of fsck in [Command Reference Manual](#).

6. *Installation of License*

The license should be installed to use the hybrid controller function for ASX/U-51 REV 6.00A and later. In [6. 1 "Installation of License" on page 6-1](#), the general procedure is explained. The software product can be used only by this procedure usually.

In [6. 2 "Management of License" on page 6-5](#), details on the license management and applications for the more complicated cases are explained.

6. 1 Installation of License

6. 1. 1 Selection of License Server

First select (decide) which of the systems will be a license server. The license server is a management system as for the license.

The most general selection is to select the system on which the product is executed as a license server. The setting is easy.

It is also available to select the license server isolated from the system on which the product is executed.

Case of floating license, the utilization rights can be obtained on any system connected by networking. Therefore, the limited utilization rights can be used effectively.

The license server software is stored when the product is installed, will be loaded into the disk automatically. That is, the system in which products are installed functions as a license server only by executing the license installation script.

6. 1. 2 Receiving the Password Registration Information

Fill out the license password issue request sheet to receive the password registration information. This subsection explains the caution that is necessary to fill the request sheet.

Sever HOST name

The HOST name of the license server. On the license server, the HOST name can be known by the following command.

```
% uname -n
```

Sever HOST ID

The HOST ID can be known by the following command under the super user of the license server.

```
# sysdef -h
```

➔ Refer to the license certification as for other items.

6. 1. 3 Installing the License on the License Server

To install the license and start the license daemon, follow the appropriate procedure below.

If you received your password by email, refer to [6. 1. 3. 1 "Registering the Password \(from E-mail\)" on page 6-2](#). If you received your password by fax, refer to [6. 1. 3. 2 "Registering the Password \(from Fax\)" on page 6-3](#).

6. 1. 3. 1 Registering the Password (from E-mail)

- 1) Save the received e-mail as file.
- 2) Starting the license installation script.

Log in to the root, specify the name of the file created at step 1), and then start the license installation script. Specify the file name with the full pathname.

```
# csh
hostname# source/at/$install
hostname# license_install -f the name of the file where the password
                           has been saved
```

(Example) hostname# license_install -f /export/home/asx/mypassmail

- 3) Answer the question on the license daemon start or not. Enter y and the return key.

The license daemon (license management software) starts.

(This question is not asked for some license types.)

```
Start daemon, OK? [Y or N] -> Y
Complete installation.
hostname# exit
# exit
```

Setting on the license server is an end. In case the license server and the product execution system are the same, all setting are complete. You can use a product.

6. 1. 3. 2 Registering the Password (from Fax)

1) Starting the license installation script.

Log in to the root and start the license installation script as shown below.

```
# csh.
hostname# source /at/$install.
hostname# license_install.
```

2) Input of license information.

When the script is started, some questions will be displayed. Input information described on the "password sheet." (Enter data according to the password sheet, although some of the items may differ depending on the license type or product version.) In case of discontinued while executing the installation script, press the return key after pressing CTRL/C.

```
Product Name          -> PADV00-00S.
D(emo) or R(egular)   -> R.
Number of Server      -> 1.
Server Host Name      -> myhost.
Server Host-ID        -> 11223344.
Expiration Date       -> dd-mm-yyyy.
C(PU fixed) or N(etwork) -> N.
Number of License     -> 10.
Password              -> abcdefghijklmnopqrst.
```

3) Answer the question on the license daemon start or not. Enter Y and the return key.

The license daemon (license management software) starts.

(This question is not asked for some license types.)

```
Start daemon, OK? [Y or N] -> Y.
Complete installation.
# exit.
```

Setting on the license server is an end. In case the license server and the product execution system are the same, all setting are complete. You can use a product.

6. 1. 4 Setting on Product Execution System (Case with Different Product Execution System and License Server)

This section explains the setting on the system on which a product is executed. The product can be used only with the copy of the license file from the license server.

It is assumed that the product has been installed on the execution system. And, do the following operation. Log in to the account in which you use the product, and copy the license file as follows.

```
% cd $AT_ASYSTEM␣
% ftp xxx␣          ←Specify the HOST name of the license server.
% Name (xxx:xxx):␣   ←Specify arbitrary account in license server HOST.
% Password:         ␣
ftp> cd /etc/opt/licenses/advantest␣
ftp> getadvantest.lic␣
ftp> quit␣
% cd␣
```

The product can be used by this.

❗ Important

When advantest.lic has already existed in \$AT_ASYSTEM and execute the above operation, the advantest.lic will be rewritten.

➔ In the case that there are two or more products and their licenses are controlled by another server; i.e., you want to make plural license files effective; refer to [6. 2. 1 "Environment Variable LM_LICENSE_FILE" on page 6-5](#).

6. 2 Management of License

Generally the software product can be used with the explanation described in previous section.

This section explains details on the license management and the method to adapt to the more complicated case.

6. 2. 1 Environment Variable LM_LICENSE_FILE

There exist two methods to transfer the license information to the software product. One is a method that the license information is placed in the specific file in the specific directory (e.g., \$AT_ASYSTEM/advantest.lic) and then the information is transferred to the software product. [6. 1 "Installation of License" on page 6-1](#) is based on this method.

Other one is a method that the license information is placed in any file in any directory. Then the file is defined to the environment variable (LM_LICENSE_FILE) and transferred to the software product. This method is very effective to adapt to various case.

When the software is executed, \$AT_ASYSTEM/advantest.lic is searched. If the license is acquired, it is used for the software. If the license is not acquired, the license file specified for the environment variable LM_LICENSE_FILE is searched.

The environment variable LM_LICENSE_FILE is specified as follows:

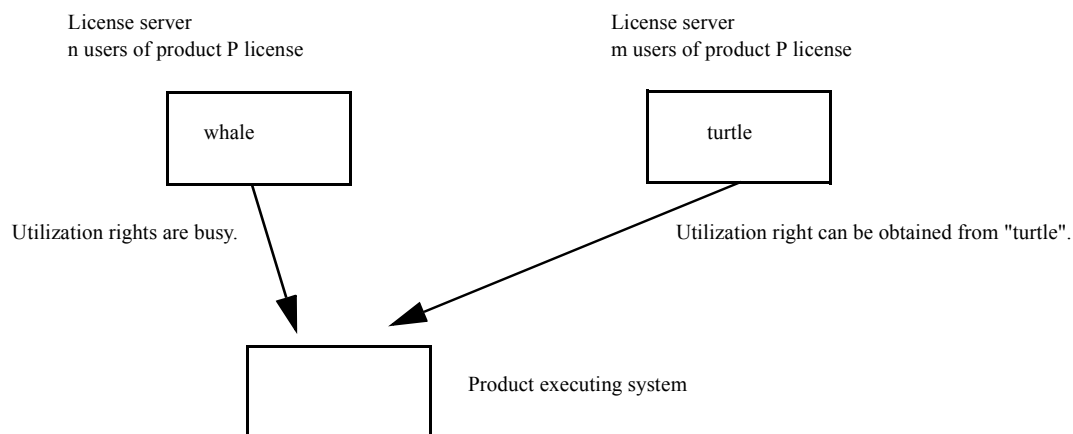
```
% setenv LM_LICENSE_FILE $AT_ASYSTEM/advantest.lic
```

If you want two or more license files to be effective, setting LM_LICENSE_FILE is needed. Plural specification becomes possible by delimiting with colon (:) and enumerating a license filename. Moreover, the case in which products except made by Advantest have already defined this LM_LICENSE_FILE is the same. It becomes effective by delimiting with (:) and adding it.

```
% setenv LM_LICENSE_FILE /usr/tool/license.dat:$AT_ASYSTEM/advantest.lic
```

6. 2. 2 Example Using Environment Variable LM_LICENSE_FILE (Using Plural Independent License Servers)

As shown in [Figure 6-1](#), it is assumed that n users and m users of the product P license are installed to license server "whale" and license server "turtle" respectively. When utilization rights of the whale server are busy, with this structure, utilization rights can be obtained automatically from the turtle server.

Figure 6-1 Using Plural Independent License Servers

In this case, make the following settings on the product executing system. (The product should always be installed to the system.)

```
% cd $AT_ASYSYSTEM
% ftp whale          ← Specify license server "whale".
Name (.. : ..):      ← Specify any account of the license server.
Password:
ftp> cd /etc/opt/licenses/advantest
ftp> get advantest.lic advantest.lic_whale
ftp> quit

% ftp turtle          ← Specify license server "turtle".
Name (.. : ..):
Password:
ftp> cd /etc/opt/licenses/advantest
ftp> get advantest.lic advantest.lic_turtle
ftp> quit

% cd
% setenv LM_LICENSE_FILE $AT_ASYSYSTEM/advantest.lic_whale:$AT_ASYSYSTEM/advantest.lic_turtle
```

6. 2. 3 Processing of Re-activation of Machine (Only License Server System)

As for the installation script `license_install` put the script (`at_lmgrd`) in `/etc/init.d` and `/etc/rc2.d` (only symbolically file) to start the license daemon automatically at the system re-starting. Therefore, you need not manually start the license daemon for re-starting the system.

6. 2. 4 Setting Redundant License Server

This is a method to manage one license password by two or more servers. The system of plural (3 systems or 5 systems) can be specified for license server. If the license daemon are active on the system of the majority, the license can be acquired. That is, if 3 redundant license server is specified and the license daemon are active on two systems, the license can be acquired. If 5 redundant license server is specified and the license daemon are active on three systems, the license can be acquired. First of all, select the license server of 3 or 5 systems. When acquisition of license is applied, specify the number of the redundant license server (3 or 5). Fill the HOST name and HOST ID of the license server that corresponds to the selected 3 or 5 systems. And, the license password is acquired. Execute the installation script of license_install for the selected machine of 3 or 5 systems as much as the independent license server.

Moreover, in the case of to which license information is changed, execute %lmutil lmdown with one system. And temporarily end the license daemon. After copying a new license file to all systems, start the license daemon. (The license daemon of all systems ends only by executing lmutil by one place.)

6. 2. 5 Start and Confirmation of the Daemon by Manual Operation

In the case of which ends only by creating the license file when installing it, or, in the case of the stop with the daemon by license management utility, the daemon is manually started.

The start is input from the command line as follows by the right of root.

- Case of sh

```
# /etc/opt/licenses/advantest/lmgrd -c
    /etc/opt/licenses/advantest/advantest.lic >
    /etc/opt/licenses/advantest/advantest.lic.log 2>&1 &␣
```

- Case of csh

```
# /etc/opt/licenses/advantest/lmgrd -c
    /etc/opt/licenses/advantest/advantest.lic >&
    /etc/opt/licenses/advantest/advantest.lic.log & ␣
```

Do not start mistaking lmgrd attached to other tools for which FLEXlm is used. Start surely with path name.

The start of two daemons can be confirmed by the ps command in the system that makes it to the license server.

```
% /usr/bin/ps -ef | egrep 'lmgrd | J-ADVANTES'␣
```

6. 2. 6 License Management Utility (lmutil)

The lmutil command is used for a temporary stop of the license daemon and the state check.

When the following are specified for the first argument, the lmutil command will offer various functions.

lmdown	Terminates the license daemon and the vendor daemon.
lmhostid	Displays host ID of the machine.
lmstat	Displays the status of the daemon and the status of the function.
lmver	Displays the version of the license management relation file.

For example, the stop of the license daemon is executed as follows.

```
% lmutil lmdown -c /etc/opt/licenses/advantest/advantest.lic
```

The name of the FLEXlm tool (for example lmutil) is almost same as the name of the FLEXlm tool of other vendor software.

Therefore, for the confirmation of the license of the product of Advantest, specify path name (/etc/opt/licenses/advantest) and start the FLEXlm tool of Advantest.

6. 2. 7 License Management Relation File and the Position

/etc/opt/licenses/advantest/J-ADVANTES	Vendor daemon
/etc/opt/licenses/advantest/license_install	Installation script
/etc/opt/licenses/advantest/lmgrd	License daemon
/etc/opt/licenses/advantest/lmutil	License management utility
/etc/init.d/at_lmgrp	Script of automatic start/end
/etc/rc2.d/S19at_lmgrp	at_lmgrp symbolic link
/etc/rc2.d/K19at_lmgrp	at_lmgrp symbolic link
/etc/opt/licenses/advantest/advantest.lic	License file
(environment variable) AT_ASYSYSTEM/advantest.lic	License file

7. *Tester Environment Settings*

This chapter describes setting items required for starting the tester controller.

When the installation of ASX/U software is completed, the account of "asx" is automatically created so that the tester controller can be used easily.

If you use the account "asx", the operation described in [7. 1 "Creating an User Account" on page 7-1](#) is not required. Go to [7. 2 "Tester Environment Settings \(ATLworks Environment\)" on page 7-3](#). If you use a user account other than "asx" to operate the tester controller, create the new user account with referring to [7. 1 "Creating an User Account" on page 7-1](#). After that, go to [7. 2 "Tester Environment Settings \(ATLworks Environment\)" on page 7-3](#).

7. 1 Creating an User Account

First create a user account. The most basic procedure to create the user account is explained below. Create the user account according to the following procedure.

- 1) Log in to root. If you log in to the root from the login screen (dtlogin) using Solaris 7 and later, the window environment is automatically started up.

If your test system is under Solaris 2.5.1, start OpenWindows as follows.

```
# /usr/openwin/bin/openwin &
```

- 2) Start the "Command Tool" or the "Terminal emulator."
- 3) Start the admintool using this Command Tool or Terminal emulator.

```
# LANG=C admintool &
```

- 4) "Admintool:Users" window appears. Then select "Edit"→"Add...".

5) "Admintool:Add User" window appears. Then set the following items. Hold other items than the following as the default.

User Name:	Enter the user name.
User ID:	Enter the user ID. It is a unique number ranged from 100 to 60,000 to identify the user.
Primary Group:	Enter the group ID. If you want it to be the same group ID as asx account, enter 999.
Login Shell:	Select C(/bin/csh).
Create Home Dir:	Confirm that the login directory creation is selected. If it is not, click here to select it.
Path:	Specify a path name of the home directory. Customary, it is specified as "/export/home/usr-name".

6) Select "OK".

7) Return to "Admintool:Users" and select "File"→"Exit" to terminate.

If you log in as a new user using Solaris 7 and later, the following menu is displayed. Use this menu to change your password.

Click **[OK]**, and enter a password according to instructions. Enter user name and the specified password again.

You need to choose an initial password for your account, or your current password has expired.

After you set your password, you must log in again using the new password.

The following dialog box used to select the default desktop environment appears.

Welcome to Solaris. The following desktop are available.
Which one would you like to use as your default desktop?

Note: You can change this default at any time by using the Session Menu located on the Desktop Login Screen's Options Menu.

Choose One:

☒ Common Desktop Environmental (CDE)

☐ OpenWindows Desktop

Common Desktop Environment (CDE) can be used as the window environment in addition to the conventional OpenWindows. The dialog box is displayed at the first-time login to the user.

Click **[OK]** to log in to the user under CDE.

7. 2 Tester Environment Settings (ATLworks Environment)

Test systems using ASX/U-51 REV 5.05 and later include ATLworks as standard. If you want to use the ATLworks environment, perform the settings described in this section.

If you use another ASX/U system, or do not use the ATLworks environment, skip to [7. 3 "Tester Environment \(without ATLworks\)" on page 7-6](#).

7. 2. 1 ATLworks Environment Settings

To use ATLworks, various settings are required. ASX/U provides a environment setup script to help simplify the setup procedure. Information that can be set by using this script is as follows.

- Setting the environment variables for the tester controller
- Setting the environment of Windows
- Defining the keyboard type on the test controller window in ATLworks
 - ➔ For information on the keypad layout, refer to [7. 4 "Layout of xterm Keypad Keys" on page 7-8](#).
- Adding "ATLworks" to menus

Set the tester environment according to the following operations.

- 1) Log in a created user account.
- 2) Execute the environment setup script.

```
% cd␣  
% source /at/$install␣  
% atlworks_setup␣
```

- 3) The question about the type of a keyboard appears.

```
* Keyboard MODEL ? (4:TYPE4 / 5:TYPE5 or TYPE6) [4/5] 5␣
```

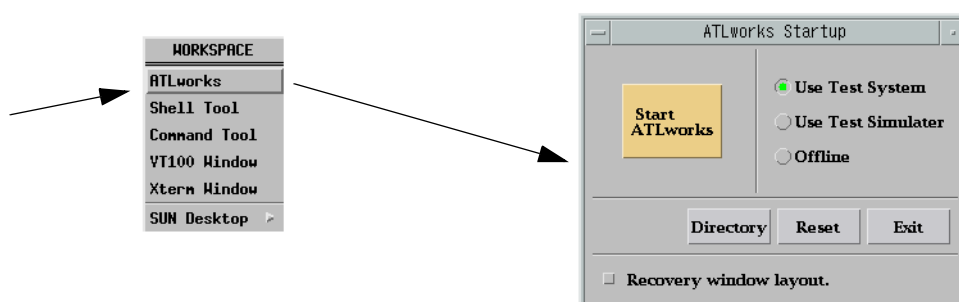
A keyboard type is printed on the middle of the bottom of the keyboard. Enter an applicable type.

- 4) Log out from the system, and log in to the system again. The window system starts and ATLworks can now be started.

7. 2. 2 Starting ATLworks

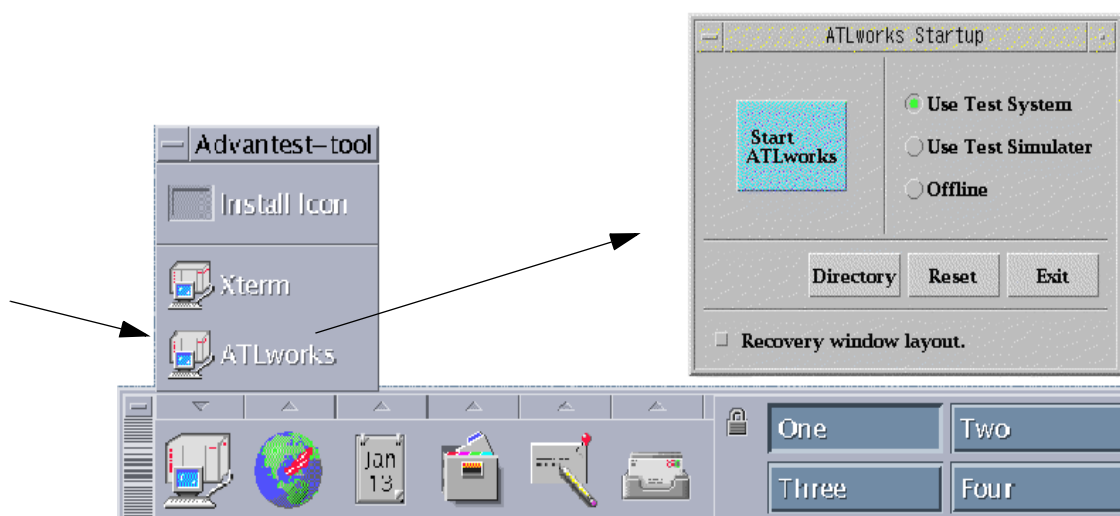
- 1) Log in to the account where the tester environment is set. The following example shows the case that the account is "asx". (The password of the account "asx" is "asx-asx" at shipping. Change the password for security. A new password should be six or more characters in length with at least one numeric character.)
- 2) When a login is done as asx, the window environment starts. The ATLworks Startup window can be opened in this environment.
 - When Solaris 2.5.1 is used, selecting ATLworks from the WORKSPACE menu displays the ATLworks Startup window.

Figure 7-1 ATLworks Startup Window (For Solaris 2.5.1)



- When Solaris 7 and later is used, the tester panel is displayed in the leftmost of the front panel. Clicking ATLworks in the tester panel menu displays the ATLworks Startup window.

Figure 7-2 ATLworks Startup Window (For Solaris 7 and Later)



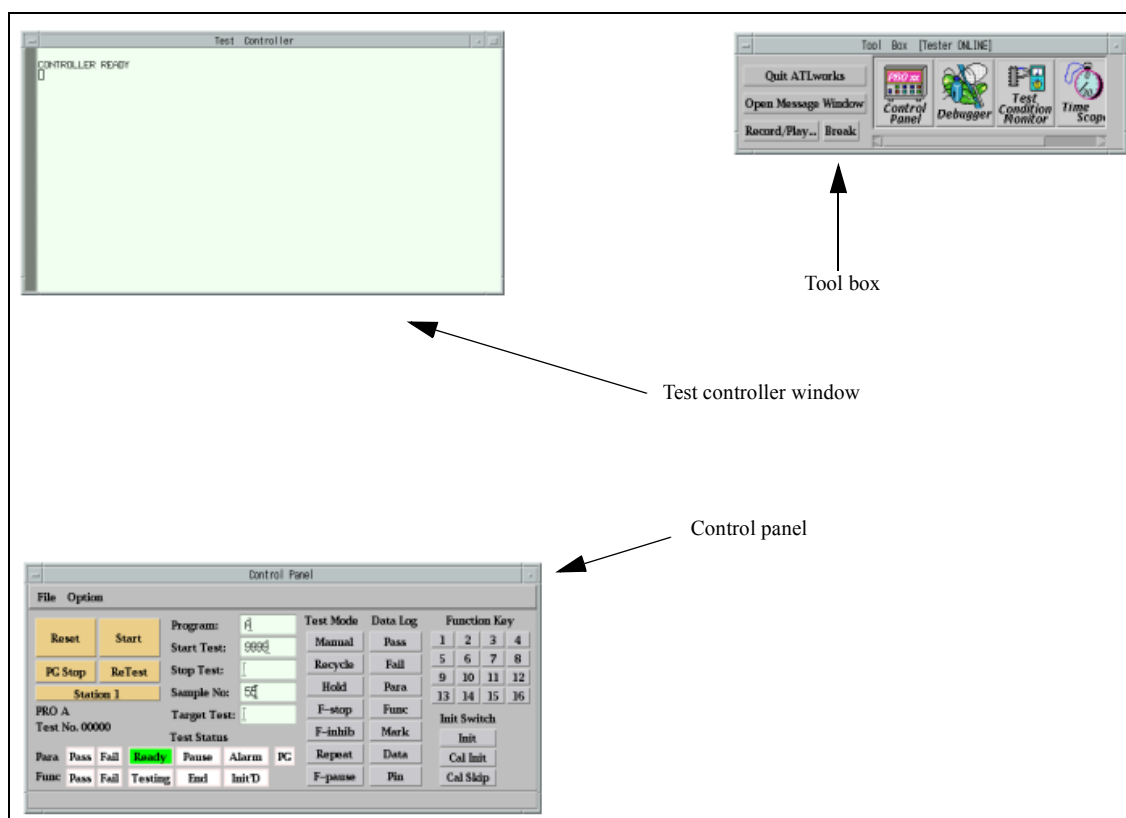
- 3) Clicking the Start ATLworks button in the ATLworks Startup window starts ATLworks.

The windows displayed after ATLworks is started are shown in [7. 2. 3 "ATLworks Initial Screen" on page 7-5](#).

7. 2. 3 ATLworks Initial Screen

Clicking the Start ATLworks button in the ATLworks Startup window starts ATLworks as shown below.

Figure 7-3 ATLworks Initial Screen



Tool box

Used to start graphic tools and to stop ATLworks.

Test controller window

Used to execute key commands and character-based utilities under the MOST environment in the test controller terminal window.

Control panel

Used to execute functions of the control box of each test station. You can specify program names, or can start and stop programs.

7. 3 Tester Environment (without ATLworks)



Tip

When Motif Window Manager is used, follow the environmental setting of each GUI tool installation procedure instead of the setting described in this section.

7. 3. 1 Tester Environment Settings

To use the tester controller, various settings are required. In ASX/U, a script is provided so that the tester environment can be set easily. Information that can be set by using this script is as follows.

- Setting the environment variables for the tester controller
- Setting the environment of Windows
- Defining the keypad for xterm
 - ➔ For information on the keypad layout, refer to [7. 4 "Layout of xterm Keypad Keys" on page 7-8.](#)
- Adding xterm to the menu
- Setting the command search pass for the online manual

Set the tester environment according to the following operations.

- 1) Log in a created user account.
- 2) Execute tester environment setting scripts.

```
% cd.
% source /at/$install.
% set-testenv.
```

- 3) The question about the type of OS appears. (This question may be omitted depending on the version used.)

```
* Japanese Solaris(SunOS) or Not? [j/n] j.
```

Enter an applicable type.

- 4) The question about the type of a keyboard appears.

```
* Type4 or Type5 keyboard ? [4/5] 5.
or
* Keyboard MODEL ? (4:TYPE4 / 5:TYPE5 or TYPE6) [4/5] 5.
```

A keyboard type is printed on the middle of the bottom of the keyboard. Enter an applicable type.

- 5) Log out once, then log in again. The window system starts and the tester controller can be executed.

7. 3. 2 Tester Controller Start and Stop

- 1) Log in to the account where the tester environment is set. The following example shows the case that the account is "asx". (The password of the account "asx" is "asx-asx" at shipping. Change the password for security. A new password should be six or more characters in length with at least one numeric character.)
- 2) When a login is done as asx, the window environment starts. Then open the VT100 emulator window (xterm). When using OpenWindows, selecting the menu can start the xterm. When CDE is used, the tester panel is displayed in the leftmost of the front panel. Clicking ATLworks in the tester panel menu can start the xterm.
- 3) Enter the command test in the VT100 emulator window. The tester controller starts.

```
% test.  
CONTROLLER READY  
When the [ESC]key is entered, the controller stops.
```

- 4) To stop the tester controller, enter the [ESC] key. The next entry is prompted.

```
%
```

! Important

Be sure to start the test controller from a VT100 emulator window (xterm,etc) or an external (VT100 or equivalent).

7. 4 Layout of xterm Keypad Keys

After setting in [7. 3 "Tester Environment \(without ATLworks\)" on page 7-6](#) completes, VT100 keypad keys in xterm are mapped as shown below.

➔ Refer to [Figure 7-4](#) and [Figure 7-5](#) for using the keypad key utility.

Figure 7-4 Type 4 Keyboard Keypad Key Layout

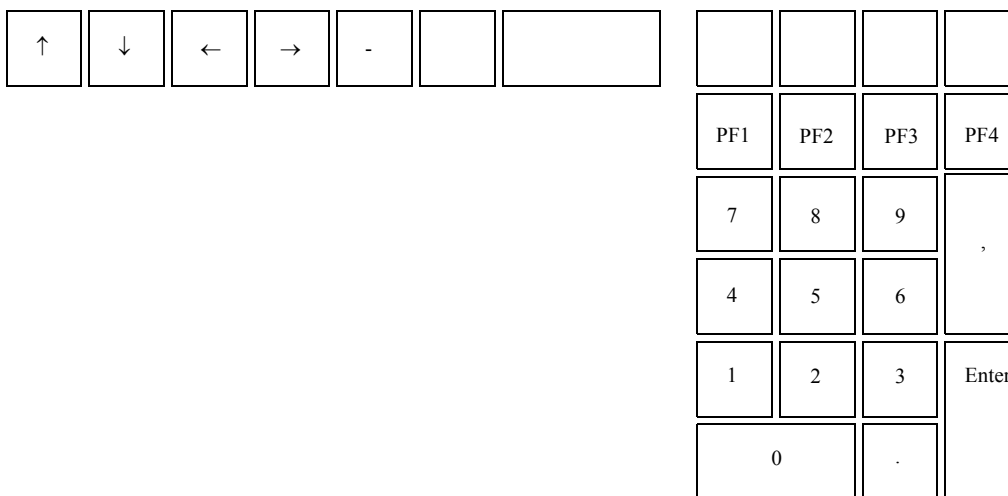
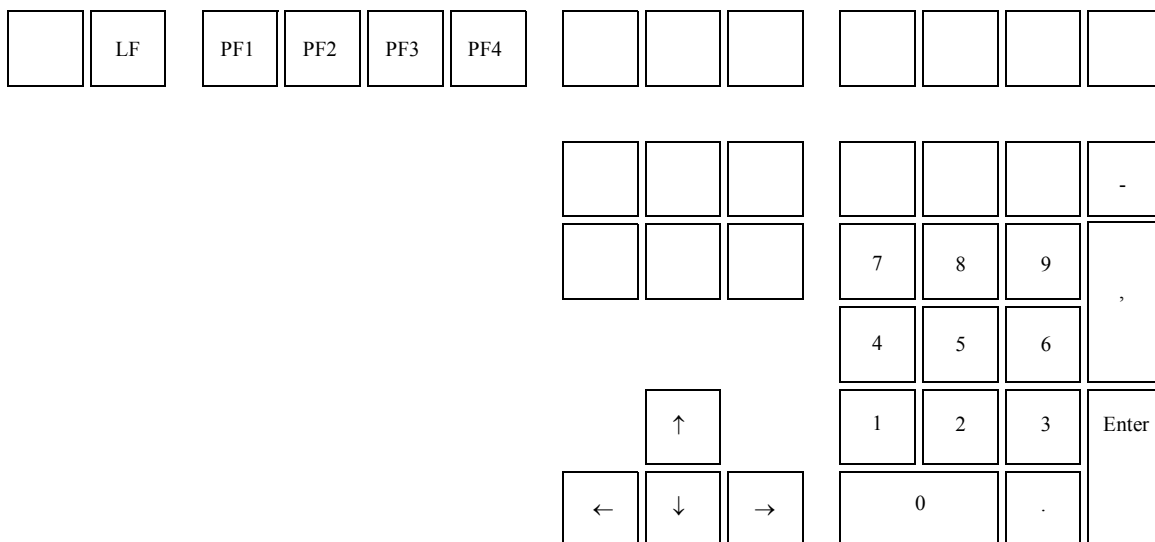


Figure 7-5 Type 5 Keyboard Keypad Key Layout



8. Installing Control Software for Optional Peripheral Equipment

This chapter explains how to install the driver software (AURAsio16) for the Aries Multiport Controller: Aries 8000P.

8. 1 Driver Software (AURAsio16) for the Aries Multiport Controller: Aries 8000P

Some software supplied with the peripheral equipment you have may not be compatible with Solaris 8. When you use the Aries Multiport Controller: Aries 8000P in the Solaris 8 environment, install the driver software (AURAsio16) that is supplied with the T series Maintenance Pack Rev 2.00 and later.

➔ For more information, refer to [T Series Maintenance Pack Rev 2.00 Release Note\(No. : 8409633\)](#).

Appendix 1. Remote System Setup for SunOS4.x.x

A remote system to be installed from a remote CD-ROM must be set up. This section explains how to set up a remote system on SunOS4.x.x.

- 1) Log in with the "root" account.
- 2) Insert a CD-ROM containing programs into the drive.
- 3) If a directory, /cdrom/cdrom0 does not exist, create it.

```
# mkdir -p /cdrom/cdrom0
```

- 4) Mount the CD-ROM.

```
# mount -rt hsfs /dev/sr0 /cdrom/cdrom0
```

- 5) Set up to make the CD-ROM available to use in other systems.

```
# exportfs i -o ro /cdrom/cdrom0
```

- 6) Start NFS mount demon. The following commands are unnecessary in a system set up as a desk server system (before this installation). Entering "ps-ax" enables to know whether the following commands are executed. If "nfsd" and "rpc.mountd" are displayed, the commands is unnecessary.

```
# nfsd 8 &
# rpc.mountd -n
```

Setting up a remote system is completed

Appendix 2. Specification to Control a Serial Port from ATL Program

This chapter explains the specification to control a serial port from an ATL program.

A2. 1 Serial Port Control

Important

This function is not used with the T5593, T5592, T5586, T5585, T5377, T5376, T5375 and T5371 test system.

A serial port of the environment processor can be controlled from the ATL program in the ASX/U environment. Specifying a file name "SYS \$CRI1" enables to input from a serial port in the same way as an usual disk file. Specifying a file name "SYS \$CRO1" also enables to output to a serial port.

[Example]

```
OPEN! CH1 R SYS $CRI1
OPEN! CH2 W SYS $CRO1
READ! CH1 TO XXX
WRITE! CH2 YYY
CLOSE! CH1
CLOSE! CH2
```

A2. 2 Specifying an Environment Variable "ADI_ACRT1"

To use a serial port in the ATL program, specify "ADI_ACRT1" as follows.

This specification can be provided by keying or ".cshrc" file of the home directory.

On the Solaris 2 and later system, setting "AT_ACRO1SPEED" and "AT_ACRI1SPEED" allow the serial port transfer rate to be specified.

A2. 2. 1 For SunOS4.x.x

The following are in case of using a port "ttya".

- Specification by keying
Execute the following commands.

```
% setenv ADI_ACRT1 /dev/ttya␣
```

- Specification by ".cshrc"
Add one line shown as follows in ".cshrc" file of the home directory.

```
setenv ADI_ACRT1 /dev/ttya      ← Addition  
exit
```

 Tip

Add the lines before "exit".

Then open the window again or enter the following command.

```
% source .cshrc␣
```

A2. 2. 2 For Solaris 2 or Solaris 7 and Later

The following are in case of using a port "term/a".

- Specification by keying
Execute the following commands.

```
% setenv ADI_ACRT1 /dev/term/a␣
```

- Specification by ".cshrc"
Add one line shown as follows in ".cshrc" file of the home directory.

```
setenv ADI_ACRT1 /dev/term/a    ← Addition  
exit
```

 Tip

Add the lines before "exit".

Then open the window again or enter the following command.

```
% source .cshrc␣
```

A2. 2. 3 Changing a Transfer Rate

When using Solaris 2 and later, serial port transfer rates can be specified.

(ASX/U-51 REV 4.03, ASX/U-30 REV 5.06A, ASX/U-73 REV 5.07, ASX/U-50 REV 6.05, ASX/U-12 REV 3.14A, or ASX/U-11E REV 2.04 and later)

Set "AT_ACRO1SPEED" and "AT_ACRI1SPEED" after "ADI_ACRT1".

Use the numbers below to set the corresponding transfer rates as shown.

13=9600 baud (default)

14=19200 baud

15=38400 baud

- Specification by keying

If setting a rate to 19200 baud, enter the following command.

```
% setenv ADI_ACRT1 /dev/term/a.
% setenv AT_ACRO1SPEED 14.
% setenv AT_ACRI1SPEED 14.
```

- Specification by ".cshrc"

Add the lines shown below in ".cshrc" file of the home directory.

```
setenv ADI_ACRT1 /dev/term/a      ← Addition
setenv AT_ACRO1SPEED 14          ← Addition
setenv AT_ACRI1SPEED 14          ← Addition
exit
```



Tip

Add the lines before "exit".

Then open the window again or enter the following command.

```
% source .cshrc.
```

A2. 3 Notice

- Only one of the following ports corresponding to each software should be chosen in this specification.

SunOS4.x.x

ttya to ttyd

Solaris 2 or Solaris 7 and later

term/a, term/b

- A port specified with "ADI_ACRT1" can not be used in other processes while "test" is running.

- If a port specified with "ADI_ACRT1" is the same as a device to start "test", it is impossible to open "SYS \$CRO1/SYS \$CRI1". Opening makes a run-time error occur.
- When a port specified with "ADI_ACRT1" has been used, (for example, with an external terminal connected and logging in), it is impossible to open "SYS \$CRO1/SYS \$CRI1". Opening makes a run-time error occur.

Change the port as follows to open "SYS \$CRO1/SYS \$CRI1".

For SunOS 4.x.x (In using a device, "ttyb".)

- 1) Become a super user and open a "/etc/ttytab" file. If a status is "on", change the status to "off".

ttyb	/usr/etc/getty std.9600	unknown	<u>on</u> local secure
		↓	
ttyb	/usr/etc/getty std.9600	unknown	<u>off</u> local secure

- 2) Close the file. Then enter the next command.

```
# kill -HUP 1
```

- 3) Change the permission.

```
# chmod 666 /dev/ttyb
```

For Solaris 2.4 (using "/dev/term/a")

- 1) Log in to the root account and start "openwindow".

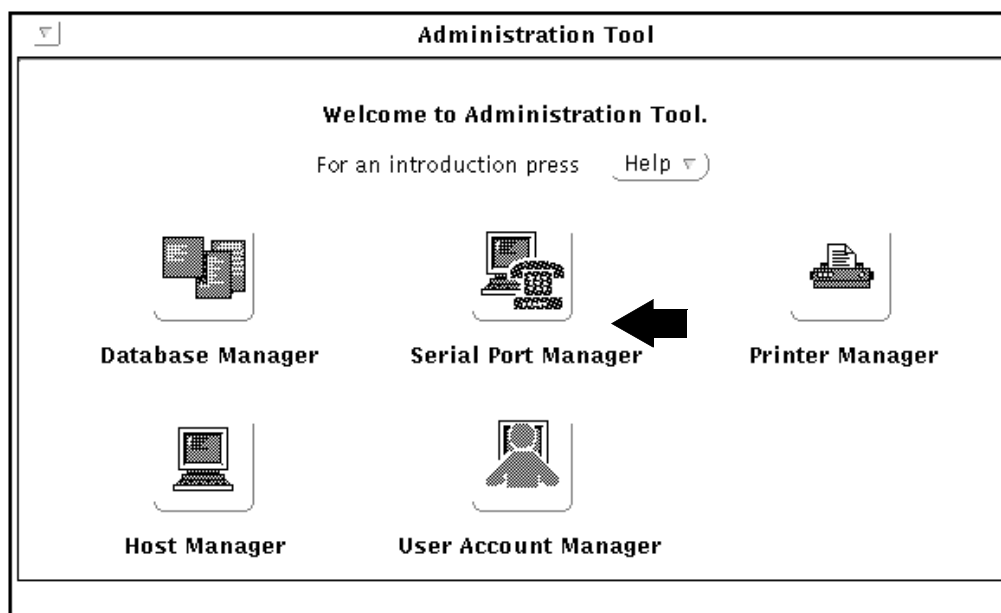
```
# /usr/openwin/bin/openwin
```

- 2) Activate "admintool".

```
# admintool
```

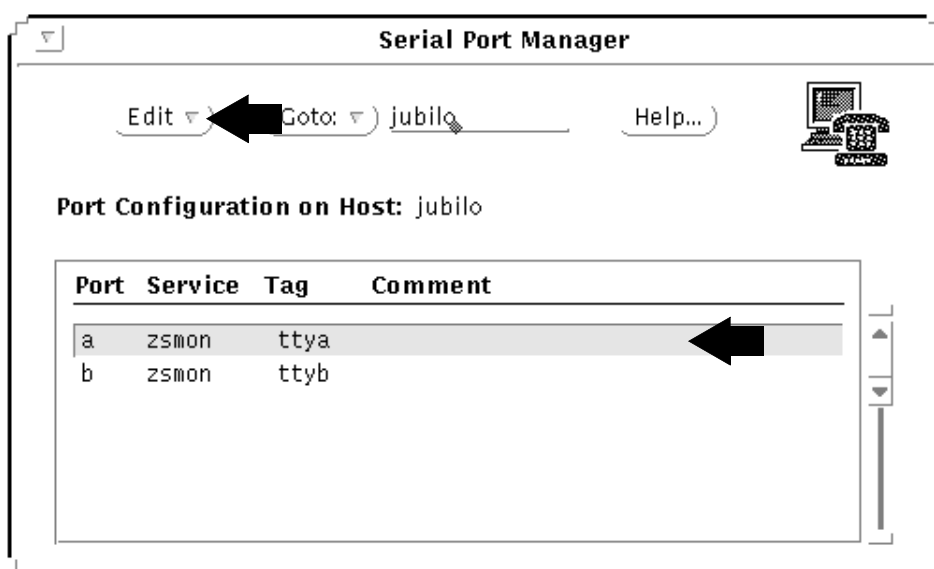
- 3) Click **[Serial Port Manager]**.

Figure A 2-1 Administration Tool



- 4) Select **[Port a]**, then click **[Edit]**.

Figure A 2-2 Serial Port Manager



- 5) The Serial Port Manager: Modify Service window appears.

Click **[Enabled]**, **[More]**, **[Use Template]** and **[Initialize Only]** in the order. Then, click **[Apply]**.

Figure A 2-3 Serial Port Manager: Modify Service

Serial Port Manager: Modify Service

Use Template ▼ 3

Detail: Basic More 2

Port: a

Service: Disabled Enabled 1

Baud Rate: ▼ 9600

Terminal type: _____

Options: ☒ Initialize Only 4
☐ Bidirectional
☒ Software Carrier

Login Prompt: login: _____

Comment: Terminal - Hardwired

Service Tag: ttya

Port Monitor Tag: ▼ zsmon

Apply 5 Reset Dismiss

Template: Terminal - Hardwired

Hostname: jubilo

After the above operations are completed, terminate "admintool".

6) Change the permission.

```
# chmod 666 /dev/term/a
```

For Solaris 2.5.1 and later (using "/dev/term/a")

1) Log in to root. If your test system is under Solaris 2.5.1, start OpenWindows as follows.

```
# /usr/openwin/bin/openwin
```

2) Start "admintool".

```
# LANG=C admintool
```

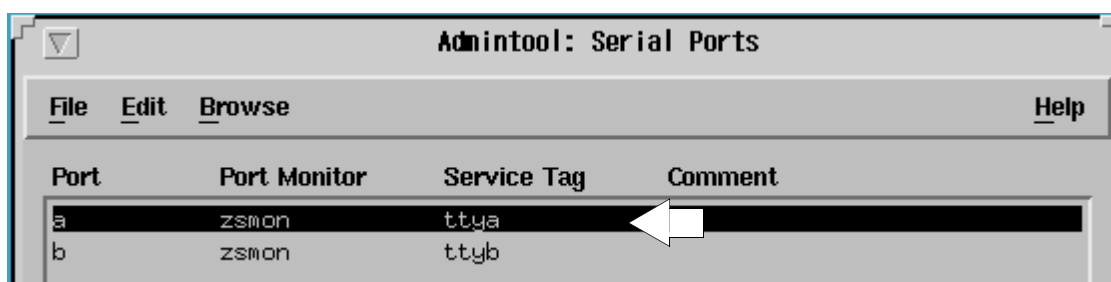
3) Click **[Browse]** and select **[Serial Ports]**.

Figure A 2-4 Admintool: Users



4) Double-click **[Port a]**.

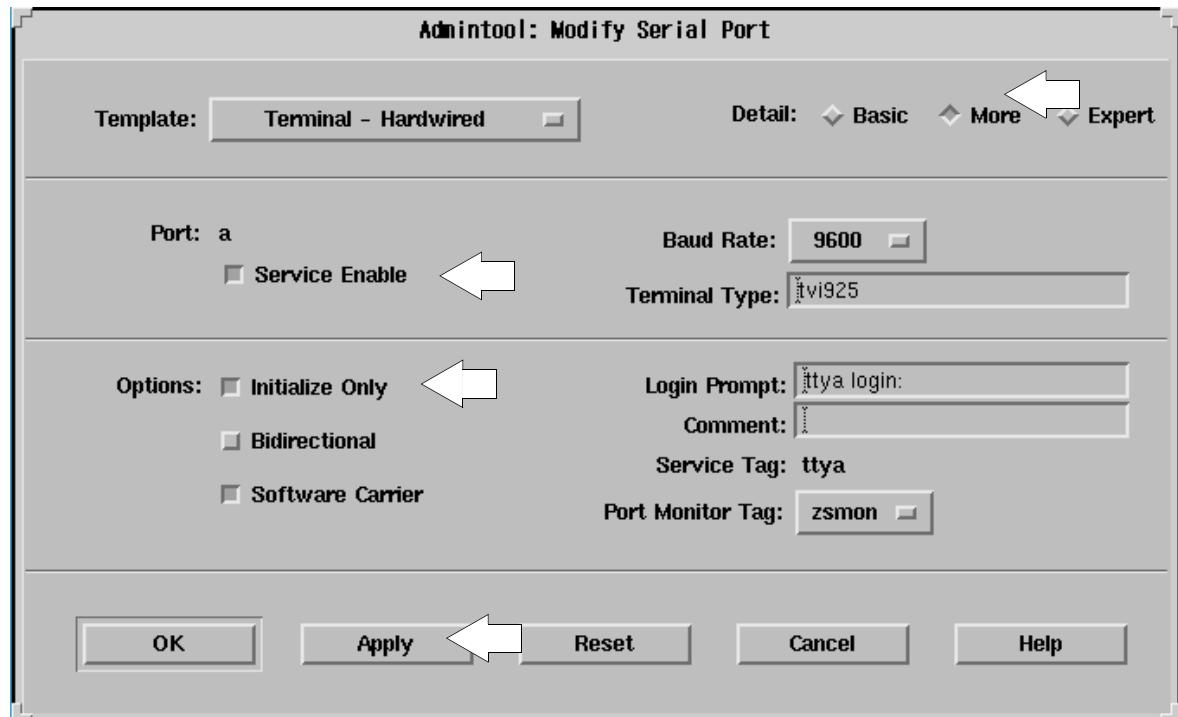
Figure A 2-5 Admintool: Serial Ports



5) Admintool: Modify Serial Port window appears.

Click **[More]** to confirm **[Service Enable]** and **[Initialize Only]** are selected. If they are not selected, make required settings and then **[Apply]**.

Figure A 2-6 Admintool: Modify Serial Port



After the above operations are completed, terminate "admintool".

6) Change the permission.

```
# chmod 666 /dev/term/a
```

Appendix 3. Changing IP Address/ hostname

When changing an IP Address or hostname of the EWS, follow instructions of a network manager. The general method of changing the IP Address and hostname is described below, but the instructions of the network manager take precedence over the method.

1) Log in to the root account. Enter the following.

```
# sys-unconfig␣  
  
...  
Do you want to continue (y/n)?y␣
```

2) After the test system terminates, enter the following command to restart it.

```
ok boot␣
```

Enter information (such as about network) according to instructions on the display.

- 3) If the host name has been changed, edit two license files using the editor as shown below. (If either file does not exist, go to the next step.)

```
# ssh.
hostname# source /at/$install.
hostname# cd $AT_ASYSTEM.
hostname# vi advantest.lic.
If the same hostname as the old one (your own hostname) has existed, you should
change your hostname.
In other case, it is not necessary to be changed.
↓
SERVER hostname ANY 7900

DAEMON J-ADVANTES /etc/opt/licenses/advantest/J-ADVANTES
FEATURE PXTL50-35S J-ADVANTES 1.000 30-nov-94 0 5B4D.....
FEATURE PTSM50-35S J-ADVANTES 1.000 30-nov-94 0 7AWS.....

hostname# cd /etc/opt/licenses/advantest.
hostname# vi advantest.lic.

If the same hostname as the old one (your own hostname) has existed, you should
change your hostname.
In other case, it is not necessary to be changed.
↓
SERVER hostname ANY 7900

DAEMON J-ADVANTES /etc/opt/licenses/advantest/J-ADVANTES
FEATURE PXTL50-35S J-ADVANTES 1.000 30-nov-94 0 5B4D.....
FEATURE PTSM50-35S J-ADVANTES 1.000 30-nov-94 0 7AWS.....
```

- 4) End with reboot.

```
# reboot.
```

Appendix 4. Lowercase and Uppercase Shift Command Entry Setting

The test controller commands should be entered in uppercase letters. Lowercase letters are not accepted. Therefore, at every time starting the controller with "%test" or every time going into the controller window from the other shell window, a case shift to the uppercase is necessary. The [Shift] key or the [Caps Lock] key is used for to enter characters in uppercase.

If the environment variable "AT_ACAPSLock" is set to on or reverse, the uppercase letters can be entered to the test controller irrespective of those operation.

To use this function, add the following line into the ".cshrc" file at the login directory.

```
setenv AT_ACAPSLock on

or

setenv AT_ACAPSLock reverse
```

This environment variable has no effect on others than the test controller. Therefore, lowercase shell commands can be used in the other window at the same condition.

If AT_ACAPSLock is set to "on," all of the entered letters are displayed in uppercase letters in the controller window.

If AT_ACAPSLock is set to "reverse," all of the entered letters are displayed in the controller window as follows.

Lowercase→Uppercase

Uppercase→Lowercase

To reset the AT_ACAPSLock setting, execute the following command and start the controller using the "%test" command.

```
% unsetenv AT_ACAPSLock
```

Important

Setting "on" in the AT_ACAPSLock environment variable is available only on or after ASX/U-12 REV 2.07A, ASX/U-30 REV 5.03B, ASX/U-50 REV 6.02C, ASX/U-51 REV 1.00 or ASX/U-73 REV 5.02 and later. It is not supported in the other systems.

Setting "reverse" in the AT_ACAPSLock environment variable is available only in ASX/U-51 REV 6.00B and later.

Appendix 5. Hints on Troubleshooting Related to the License

This chapter describes hints of troubleshooting and the contents of error messages.

A5. 1 Hints on Troubleshooting

- Check the log file. Various kinds of information are recorded in it.
(On the license server)
Log file: /etc/opt/licenses/advantest/advantest.lic.log
- Execute the following commands. The condition of the license daemon will be displayed.
(On the license server)
/etc/opt/licenses/advantest/lmutil lmstat -a -c /etc/opt/licenses/advantest/advantest.lic
- When a problem occurred after going through various circumstances, it may be wise of you to clear then reset all license information. All license information can be cleared by the following procedure. Log in to the root account.

```
# csh.␣
hostname# source /at/$install.␣
hostname# cd $AT_ASYSTEM.␣
hostname# rm advantest.lic.␣      ←Delete the license file if existing.
hostname# rm /etc/opt/licenses/advantest/advantest.lic.␣
                               ↑Delete the license file if existing.
hostname# reboot.␣              ←Restart the system.
```

- After the new license is added with `license_install`, the licenses that could have been used before the addition become unavailable.
When the license file is specified with the environment variable "LM_LICENSE_FILE", care is needed in adding license.
When `license_install` is executed, `$AT_ASYSTEM/advantest.lic` is created. Then, the Advantest-product comes not to refer to LM_LICENSE_FILE.
Delete `$AT_ASYSTEM/advantest.lic` and make a setting of LM_LICENSE_FILE with referring to [6. 2. 1 "Environment Variable LM_LICENSE_FILE" on page 6-5](#) and [6. 2. 2 "Example Using Environment Variable LM_LICENSE_FILE \(Using Plural Independent License Servers\)" on page 6-5](#).

A5. 2 Error Messages

(On executing license_install)

Inputted server name(s) do not match with exist sever name(s).

Do you want to overwrite the server name(s)
in advantest.lic ? [Y or Q(quit)]->

The license cannot be installed because the entered host name is different from the host name of the already-registered license. It is supposed that the demo license without host name specification has already been installed. If you may change the host name in the advantest.lic file, enter "Y" or "y". The installation process will continue. If you don't want to change the host name, enter "Q" or "q". The license installation will quit.

Maximum # users for "....." reached.

The product cannot be used on exceeding allowable number of users.

Wrong host for

It is executed on the system having no CPU specification license.

client: feature has expired

The feature has expired.

Checkout of "....." failed.

client: cannot connect to license server (Connection refused)

It cannot be connected to the license server. Check file "/etc/opt/licenses/advantest/advantest.lic.log" on the license server.

"(lmgrd) Retrying socket bind (address in use: port 7900)" has been recorded, the port number is being used in the other process. The following two cases are supposed.

1. After the license is changed with license_install, on the system operated as a license server, the previously-used port may not be released. Then wait about five minutes. The previously-used port is released and the port "7900" becomes available.
2. The port "7900" may be used in the other tool. In this case, change the port number. That is, change 7900 in "SERVER hostname hostid 7900" in the license file defined to "/etc/opt/licenses/advantest/advantest.lic", "\$AT_ASYSTEM/advantest.lic" and environment variable "LM_LICENSE_FILE" to other number without duplication. Then, re-execute them. In this case, be careful that the port number is not duplicate with other by referring to /etc/services.

License: All license(s) for "....." are currently in use
AND QUEUED the request

As all of permissible network licenses have already been used, the system issues a request of use and is queuing. The system starts operation when it get a license. If there is no need to wait, quit with CTRL/C. (Only XATL has the function of waiting in order to get a license.)

CAUTION!!! This Feature ("Product - Name") expires at : date

Licenses with expiration such as the demo license are installed. Note that they cannot be used after the expiration date.

Appendix 6. Testers Start and Stop for Both MOST and ASX/U

This chapter explains how to start and stop the environment processor and the tester processor.

A6. 1 Tester Processor Start and Stop

Start

- 1) Set the DEVICE SELECT switch ((1) in [Figure A 6-1](#) and [Figure A 6-2](#)) on the AC control panel to 15.
- 2) Turn on the POWER switch ((2) of [Figure A 6-1](#) and [Figure A 6-2](#)).

Stop

- 1) Push the HALT button ((3) in [Figure A 6-1](#) and [Figure A 6-2](#)) of AC control panel.
ALARM lamp is on and RUN lamp goes off.
- 2) Turn off the POWER switch.

Figure A 6-1 Tester Processor AC Control Panel

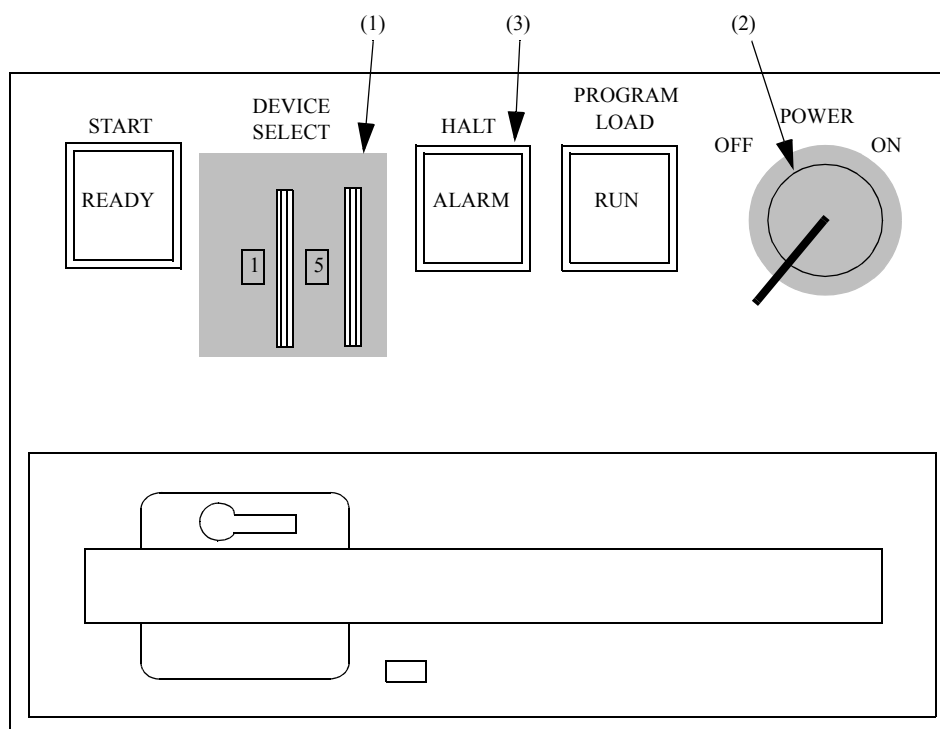
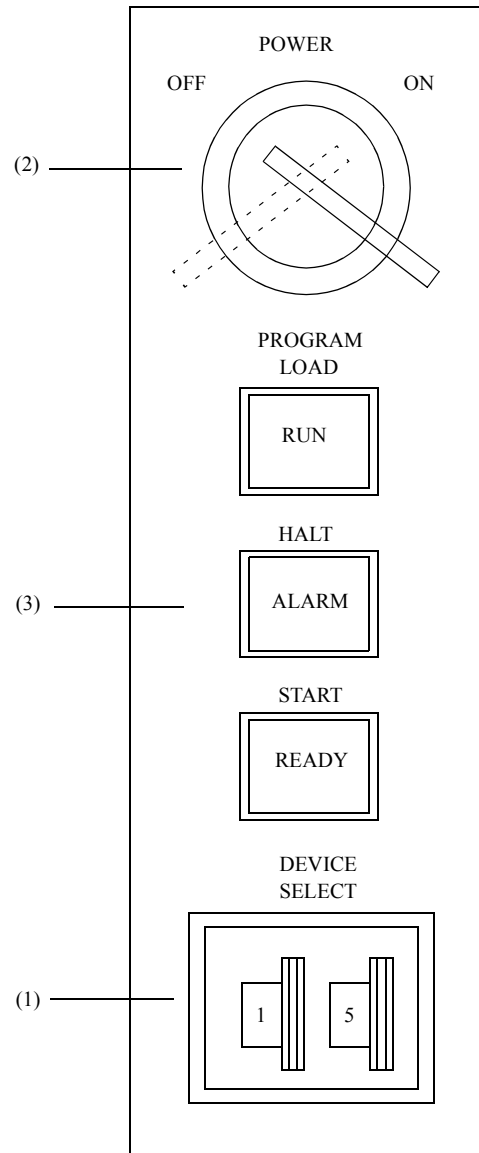


Figure A 6-2 Tester Processor AC Control Panel (cont'd)

A6. 2 Environment Processor Start and Stop

Start

- 1) Check that the tester processor is powered up.

- 2) Turn on the power switch in the display. Next, turn on the power switch of the environment processor. POWER ON lamp lights up.
- 3) Many kinds of starting information are scrolled on the screen and finally the scrolled following display appears, and finally the following message is displayed.

```
hostname login:
```

A login dialog box (dtlogin) is displayed by Solaris 7 through Graphical User Interface (GUI). Enter the user name and password to log in the system as well as the conventional login.

- 4) The system start is then completed and the system enters the status where log in is enabled.

! Important

The following accounts are registered in ADVANsite at shipping.

Account	Password
root	None
asx	asx-asx

Set the password or change the password for security.

Stop

- 1) Log in to root.
- 2) Enter the shutdown command.
 - For SunOS 4.x

```
# shutdown -h now␣
```

- For Solaris 2 or Solaris 7 and later

```
# /usr/sbin/shutdown -y -i0 -g0␣
```

- 3) The prompt, ">" or "ok" are displayed on the screen and the environment processor stop processing is completed:

```
shutdown at 17:01 (in 0 minutes) [pid 142]
#

*** FINAL System shutdown message from root@sun ***

System going down IMMEDIATELY

System shutdown time has arrived
Dec 13 17:01:16 sun shutdown : halt by root
Dec 13 17:01:18 sun syslogd : going down on signal 15
syncing file systems... done
Halted

> __←Cursor   or   ok__←Cursor
```

- 4) Turn off the processor power switch. Then, turn off the display power switch.

A6. 3 Emergency Stop

Turn off the processor power switch. Next, turn off the display power switch.

! Important

When the above emergency stop is performed, the file system may be damaged. When the system is reactivated, the command fsck (file system check) recovers the damaged part automatically as much as possible. However, if the file system has been badly damaged, the command fsck might ask the operator who performs installation for details of the processing.

➔ For details, refer to the explanation of fsck in [Command Reference Manual](#).

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